

Project Design Phase-II
Technology Stack (Architecture & Stack)

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Project Name	ShopEZ
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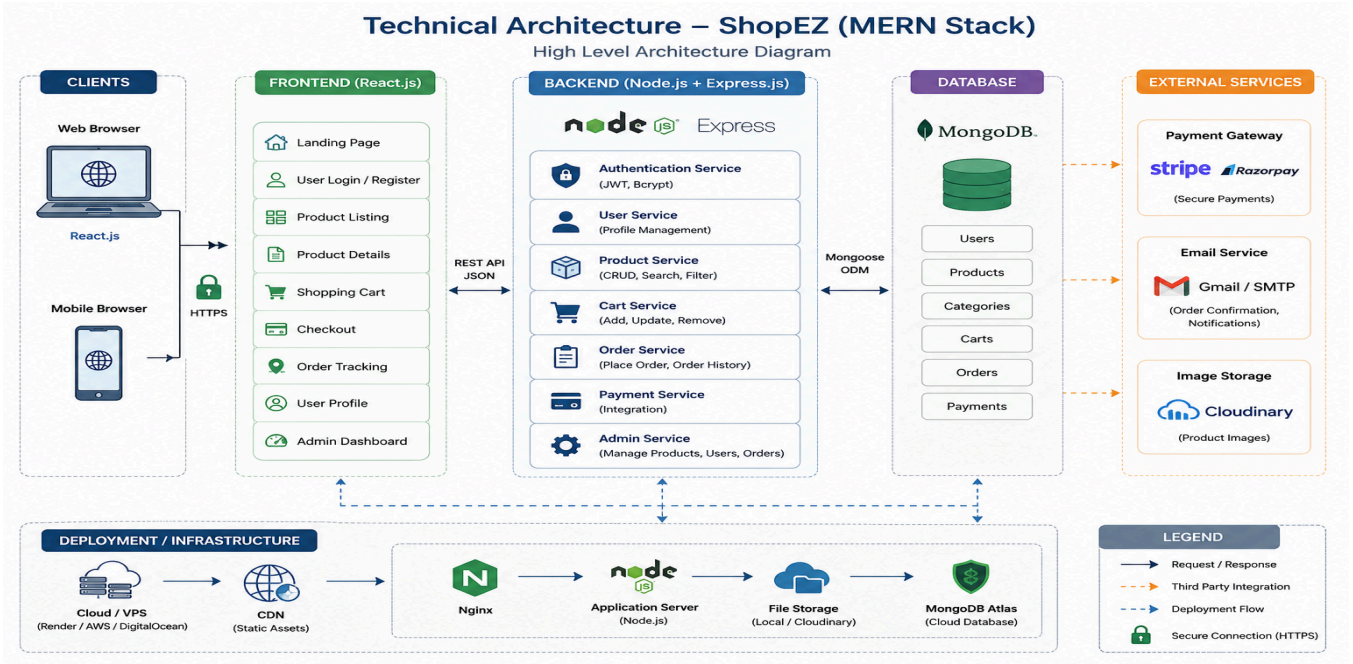


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web application for customers and administrators	HTML5, CSS3, JavaScript, React.js, Bootstrap
2	Application Logic-1	User Authentication & Authorization	Node.js, Express.js, JWT, bcrypt
3	Application Logic-2	Product, Cart and Order Management	Express.js REST APIs, Node.js
4	Application Logic-3	Admin Dashboard, Analytics & Product Management	React.js, Express.js
5	Database	Stores users, products, categories, carts and orders	MongoDB, Mongoose ODM
6	Cloud Database	Database hosting (Future Deployment)	MongoDB Atlas
7	File Storage	Product images and static assets	Local Storage / Cloudinary (Optional)
8	External API-1	Payment Processing	Stripe API / Razorpay API (Future Integration)
9	External API-2	Email Notifications	Nodemailer / Gmail SMTP
10	Machine Learning Model	Product Recommendation System (Future Enhancement)	Recommendation Engine (Optional)
11	Infrastructure (Server / Cloud)	Application Deployment	Localhost, Render, Vercel, MongoDB Atlas

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source frameworks used for frontend and backend development	React.js, Node.js, Express.js, Bootstrap
2	Security Implementations	Secure authentication, password encryption, protected APIs	JWT Authentication, bcrypt, HTTPS, CORS
3	Scalable Architecture	Three-tier MERN architecture with REST APIs supporting future expansion	MERN Stack, MVC Architecture
4	Availability	High availability through cloud deployment and reliable database services	MongoDB Atlas, Render, Vercel
5	Performance	Fast API responses, optimized database queries, efficient frontend rendering	React Virtual DOM, Express.js, MongoDB Indexing